

Ishika Bhaumik

bhaumikiman26@gmail.com | +1 (530) 760-8800 | linkedin.com/in/ishika-bhaumik | Portfolio

Education

University of California, Davis, M.S. in Computer Science
KIIT University, B.Tech in Computer Science

Sep 2024 – Jun 2026
Jun 2018 – May 2022

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, C++
Frontend: React, Next.js, React Native, Tailwind CSS, Component Architecture, State Management
Backend: Node.js, FastAPI, Spring Boot, REST APIs, GraphQL, Microservices, API Design
AI & Agents: LLM APIs, Agentic Workflows, RAG, Prompt Engineering, Context Engineering, Tool Calling, Structured Outputs, Evals, AI-assisted Development (Claude Code, Cursor)
Cloud & Infra: AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD Pipelines, PostgreSQL, Redis
Tools & Practices: Git, Agile, Code Reviews, Unit & Integration Testing, Technical Documentation

Experience

Associate Software Engineer, Novartis, Hyderabad, India Aug 2022 – Aug 2024

- Built and shipped **full-stack features** end-to-end using **React, TypeScript, Java, and REST APIs** — owned architecture, implementation, deployment, and production support for applications serving **4,000+ users**.
- Designed and integrated **third-party APIs and backend services**, enabling new automated workflows and eliminating manual processes; worked directly with stakeholders to translate real-world workflows into production-grade software.
- Optimized frontend-backend interactions and **SQL query performance** through systematic profiling, reducing system latency by **30%**; debugged production issues end-to-end using structured logging and monitoring.
- Collaborated cross-functionally with product, design, and engineering teams; contributed to **code reviews** and technical documentation to raise team-wide quality standards.

Software Engineering Intern, Novartis, Hyderabad, India Jan 2022 – Jun 2022

- Contributed to production **full-stack features** using **React, Java, and Spring Boot**; implemented and tested **REST API** endpoints shipped to real users.
- Participated in **code reviews and Agile sprints**, building habits around reliability, testing, and iterative delivery.

Software Engineering Intern, HighRadius Technologies, Hyderabad, India Jun 2021 – Dec 2021

- Built **React-based dashboards** for financial data workflows integrated with **Snowflake and SQL pipelines**, supporting analytics for 50+ users.
- Optimized backend queries and data processing pipelines, reducing latency by **50%**.

Projects

SafeWalk — AI Safety Companion — *React Native, TypeScript, Node.js, LLM APIs, WebSockets, Google Maps API* 2026

- Built an **end-to-end agentic application** — designed the agent architecture, integrated **LLM reasoning** with real-time data sources (Google Maps, location APIs), and orchestrated multiple external services into a reliable, low-latency pipeline with **structured outputs**.

WaitWhat.ai — *Python, TypeScript, Gemini API* 2025

- Designed an **LLM-powered agentic workflow** using structured prompting, output validation, and schema-conformant results, built with correctness and auditability in mind to reliably process unstructured inputs into actionable insights.

Gas-Fee Mechanism in ResilientDB — *C++, GraphQL* 2024

- Designed and implemented a resource governance mechanism for a **Byzantine Fault Tolerant distributed database** in **C++**, ensured transaction integrity and optimized concurrent performance under high-throughput load across a multi-node environment.

Trending Tech Stack Analyzer — *React, Python, Spring Boot, PostgreSQL, AWS* 2024

- Built a **full-stack data pipeline and visualization platform**, ingested, processed, and surfaced structured insights from large unstructured datasets; deployed on **AWS** (EC2, S3, Lambda) with CI/CD pipelines.

Leadership & Achievements

- Winner, Best AI Depth & Integration — HackHayward 2026** — SafeWalk recognized for agent architecture quality, reliability of cross-system integration, and depth of AI integration under constraints.
- Best Innovation Project**, Novartis — end-to-end automation system delivering **96%** reduction in reporting time; identified opportunity, designed pipeline, and shipped independently.
- Mentored students through coding labs, debugging workshops, and code reviews as **Teaching Assistant** at UC Davis.